

DynamiX Device Reference

Every device, every knob, what it means

Generated from the live device registry

September 25, 2026 · kernel figures drawn by `make_figures.py` from the app’s own kernel code

How to read this document

A *device* is one processing tool in a layer’s chain. **Transforms** compute (they head chains and produce a result bundle); **filters** select from a computed result (list indexes and masks — scrubbing a filter knob redraws without recomputing). Chains are always transforms-then-filters; interleaving is rejected at construction.

Conventions that hold everywhere:

- **Scales are stored in pixels**, converted only for display.
- **The wavelet family is a property of the method**, never an orthogonal knob: each tool’s menu lists only the kernel class its method admits, under the method’s own names.
- **Inert knobs**: a parameter that binds only under one choice of another parameter (e.g. q only for q-family wavelets) is ignored otherwise; where the device declares it, the knob is *greyed out* in the device box while it does not apply (its value is kept).
- **View-only knobs** (marked *view* in the tables) choose what is shown from a result that is already computed and cached — Show, Component, Group, Orientation, Show PC. Switching one redraws at once; it never recomputes.
- **Orientations are (angle, frame)** with `NorthRef` $\in \{\text{grid, true, magnetic}\}$ — never a bare float.
- **No-support masking**: pixels with no honest signal support (nodata flats, FFT dust) are NaN in every exponent map — absence, never fabrication.
- Each table lists: parameter (UI label), default, hard range, and meaning. Soft ranges (knob-drag limits) are narrower than hard ranges.

Wavelet normalization: L^1 , not L^2

Every wavelet in the app is normalized in the L^1 (pure-dilation) convention, $\psi_a(x) = a^{-d} \psi(x/a)$ in $d = 2$ dimensions, so that at a Hölder singularity the transform scales as $|T_\psi[f](x, a)| \sim a^{h(x)}$ with no offset. The L^2 convention ($a^{-d/2}$, unit energy $\int \psi^2 = 1$ — e.g. the normalization constants A and A_q of Borges et al. 2004, whose transform carries $a^{-1/2}$) shifts every exponent by $d/2$. Turiel’s microcanonical formalism uses the same L^1 dilation $\Psi_r(x) = r^{-d} \Psi(x/r)$ (Turiel et al. 2008, eqs. 6–8). Borges et al.’s own WTMM example agrees in practice: their eq. 41 reads $T \sim a^\alpha$ (their fig. 5 recovers $\alpha = 0.5$), which holds for L^1 scaling, not for the $a^{-1/2}$ of their eq. 10.

- **wtmm2d normalizes in the Fourier domain**. Each filter is built as $\hat{\psi}(a\mathbf{k})$ from a smoothing function with $\hat{\theta}(0) = 1$ (unit mass at every scale), times the derivative factor $i a\mathbf{k}$: scale enters only through $a\mathbf{k}$, which *is* the L^1 dilation.
- **The Hölder tools normalize in real space** by default, on the pixel grid: positive kernels (`holder_measure`) to unit discrete mass; Mexican hats (`holder_multiaffine`) to exact zero mean, then unit L^1 norm

($\sum |\psi_r| = 1$), at every scale. With *Wavelet space* set to Fourier (§5) they are built as $\hat{\psi}(r\mathbf{k})$ instead, like `wtmm2d`'s filters.

- A constant prefactor that does not depend on the scale (such as A or A_q) cannot change any exponent — it only shifts $\log |T|$ — and the app's own normalization replaces it anyway. Only the *scale dependence* of the normalization matters.
- **Validated on a synthetic cusp** $|x - x_0|^h$: both routes recover the slope h (tests pin it; an L^2 slip would read $h + 1$ in 2-D). The smallest scales are biased by the discretization of both kernel and signal: at a sampled cusp the slope settles above $\sigma \approx 5$ px (`wtmm2d`) and $r \approx 10$ px (the Mexican hats), later for sharp-cutoff kernels ($q_T < 1$).
- **Two different q 's.** q_T (the knob *Tsallis* q of the Hölder tools) is the Tsallis/Borges shape parameter of a kernel; q in $Z(q, a)$, $\tau(q)$ and $D(h)$ is the moment order of the partition functions. They are unrelated.

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1 The workspace: datasets, layers, ROIs, derivatives

Datasets. Every opened file is a *dataset*: one row in the layer tree, and that row *is* its raw layer (its H hides the raster; right-click → Remove dataset takes the dataset and everything on it). A plain image (PNG/JPEG)

opens as one luminance band; a multiband GeoTIFF / netCDF / HDF stack opens as (n_y, n_x, n_c) .

Children. A tool dropped on a dataset never replaces the data: it spawns a *child* layer holding that tool, nested under the dataset. A child re-runs its chain on its parent’s data whenever that changes. Right-click on a result row: *Delete layer* (its children move up to its parent) or *Delete layer and children*; shift-click or Cmd-click (Ctrl on Windows) selects several rows for one Delete. **H never cascades:** each row hides only its own drawing (the dataset’s H hides the dataset raster, not the results over it).

ROIs. A saved region of interest is stored in whole *file* pixels and shown as a row **dataset @A** under its dataset. Selecting the ROI row makes it the region the next tool runs on; results (**tool @A**) nest under it. The ROI row’s H hides only its outline. An ROI cannot be deleted while results use it (delete them first). Analyzers read their own declared margin around the ROI; decompositions (**pca**, **tucker_H00I_H0SVD**, **ssa2d**) analyse the ROI itself (margin 0: a margin would mix outside data into the statistic).

Progress is live: every compute reports one increasing fraction from inside its own loops.

Derivative datasets (Fork)

A *child* keeps re-processing its parent. A **derivative** is forked once from what a result shows and from then on is an *original raw dataset*: nothing links it back, and a tool dropped on it spawns a child that analyses the derivative’s own pixels.

Where	result row → right-click → <i>Fork derivative dataset...</i> ; or the <i>Fork derivative...</i> button of the Components window
Raster	a checklist of every raster the result holds: the one on screen first (after its view knobs — e.g. Group “1”, Show residual = data – C1), then the components / PCs, recon and residual per input band of a multiband field, filtered, $h(x)$...; every checked raster becomes a band (n checked → an n -band dataset)
Vector	the extrema and maxima lines the result shows (after its filters)
Both	raster and vector go in one file
Placement	a dataset row of its own, or nested inside the dataset it came from (removing that dataset lifts the derivative to the top)
Keep	<i>Save to a file...</i> (a Save-As dialog; permanent), or <i>Temporary</i> : a file in this session’s scratch folder, left out of a saved project (a notice names it) until <i>Save derivative as...</i> on its row writes it where you choose
Grid	the result’s own grid, never resampled: the whole field, or the ROI window for an ROI result
Georeference	the parent’s CRS rides over (with the grid’s axes); the parent file’s layout (its source path, window, full size) does not
File	a RasterField .npz (values, axes, frame, units, provenance) — it opens like any .npz ; vectors ride under vec.* keys

A derivative that carries vectors gets `derived_vectors` (a loader the shell places, in the browser's Dev group) as its first step: its frozen extrema and maxima lines draw like any result's, and the extrema and chain filters can be dropped after it.

The Components window

Right panel → Decomposition → *Components...*, on a `ssa2d` or `tucker_HOOI_HOSVD` result:

- **thumbnails** — the eigenarrays (`ssa2d`: the $L_r \times L_c$ window patterns) or the component images, each labelled with its share;
- **the shares** on a log scale (group members in amber) — a slowly decaying floor is noise;
- **the w-correlation matrix** (`ssa2d`; black 0 to white 1) — separable components are nearly w-orthogonal, so groups show as blocks on the diagonal; hovering reads $w(C_i, C_j)$.

Clicking thumbnails picks the **group**: click one, Cmd-click (Ctrl) to add or remove, shift-click for a range, *All* for every one. The pick is written into the tool's Group knob (and the window follows the knob), so the canvas redraws from the cache.

2 Field preparation

2.1 noise — dither / perturbation

Adds controlled noise to the field before analysis. Its main use is *dithering* quantized data ($\pm \frac{1}{2}$ LSB) so that quantization plateaus do not read as zero-gradient flats.

Parameter	Default	Range	Meaning
<code>dist</code> (Distribution)	uniform	uniform / gaussian	noise law
<code>amplitude</code>	0	$[0, 10^9]$	0 = auto $\frac{1}{2}$ LSB; uniform: half-width; gaussian: σ
<code>seed</code>	0	$[0, 2^{31}]$	RNG seed (deterministic reruns)
<code>sign</code> (Apply)	add	add / subtract	subtract undoes a previous add exactly

3 Primary analyzers

Each primary analyzer is its own representation of the field. Dropping a *different* primary analyzer onto an analyzed layer forks a sibling layer rather than mutating the chain; consumers (filters) follow their producer on forks.

3.1 wtmm2d — canonical WTMM (Arnéodo)

The canonical multifractal machinery: a continuous wavelet transform over a geometric scale grid, modulus *maxima* detected per scale (along-gradient NMS), maxima chained across scales into maxima lines, feeding partition functions $\rightarrow \tau(q) \rightarrow$ Legendre $D(h)$. The maxima are dense by design — they are the statistical support, not a lineament map (use the filters, or the other analyzers, for sparse features). Legendre spectra are concave hulls by construction; the microcanonical tools (§5) measure $D(h)$ directly — together they diagnose non-concavity.

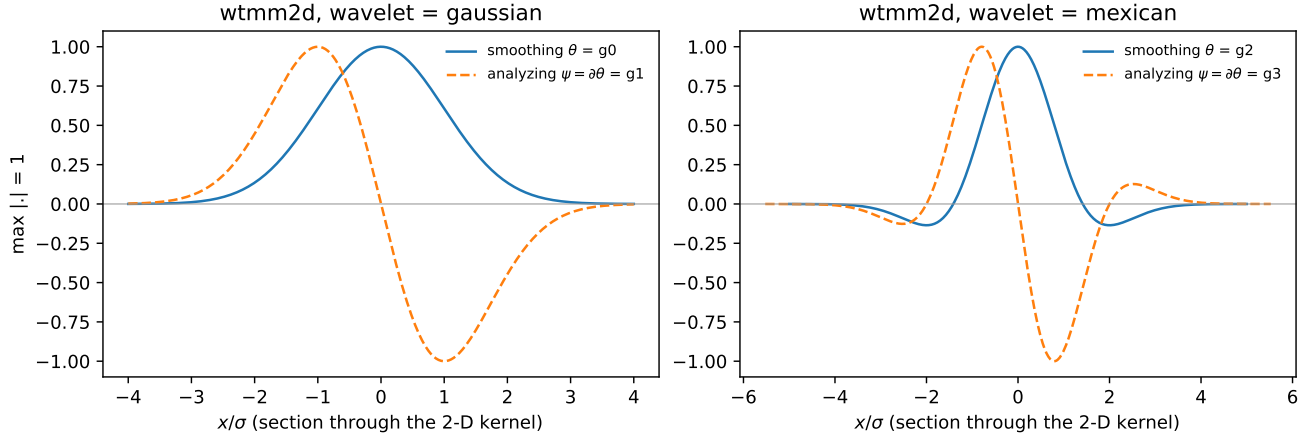


Figure 1: **wtmm2d**: the smoothing function θ and the analyzing wavelet $\psi = \partial\theta$ on the same axes (sections through the 2-D kernels, x in units of the smoothing σ). The wavelet is the derivative of the smoothing function, so the modulus maxima of ψ are the edges of θ -smoothed data.

Parameter	Default	Range	Meaning
n_oct (Octaves)	3	[1, 8]	scale range: a spans n_oct octaves above $a_{\min}$
n_voice	4	[1, 16]	scales per octave (grid density)
a_min ($a_{\min}$)	1.0	[0.25, 16] px	finest analyzed scale
wavelet	mexican	mexican / gaussian	CWT kernel (Mexican hat = LoG; gaussian = gradient pair)
dither	off	bool	$\pm \frac{1}{2}$ LSB dither for quantized rasters
fracint_alpha (η)	1.0	[-2, 3]	per-scale a^η lift (pseudo fractional integration; shifts effective h by η)
min_chain_len	2	[2, 64]	discard maxima lines spanning fewer scales
smooth	on	bool	smooth maxima-line traces
thresh	0.001	[0, 1]	modulus floor for maxima detection (fraction)
dist2_max	50	[1, 500] px ²	max squared jump when chaining maxima across scales
box_ratio	1.0	[0.1, 8]	chaining search-box anisotropy
similitude	0.8	[0, 1]	cross-scale similarity acceptance for chaining

3.2 wtmm2d.roi — WTMM over one ROI window

The parent’s WTMM re-pointed at a native-resolution window: same knobs as **wtmm2d** (minus the fractional-integration lift) plus the window itself. Used by the overview drill-down: draw an ROI on the decimated whole-extent, compute native.

Parameter	Default	Range	Meaning
roi_row, roi_col	0	px	window origin (native pixels)
roi_h, roi_w	512	≥ 8 px	window size
boundary	auto	auto / reflective	boundary handling at the window edge

3.3 mz_edges — Mallat–Zhong dyadic multiscale edges

“Discrete Approximation of Multiscale Edges” (Mallat–Zhong 1992) as a peer analyzer: the à-trous dyadic transform on the mirrored torus, per-level gradient-pair maxima (the multiscale Canny), wrap-merged into per-scale H-lines (**line_id**). Stored moduli are the paper’s own λ -normalized values (Table II). Scales are dyadic, 2^{k+1} px.

The `frac_bspline` wavelet fractionalizes the refinement order (Unser–Blu): smoothness order α becomes real, one vanishing moment always (maxima-are-edges *is* the method); $\alpha = 3$ is the paper’s wavelet exactly, with numeric λ_j^α from the rediscovered Table-II definition (discrete step-edge peak over $\theta_\alpha(0)$; $\theta_\alpha(0) = \frac{1}{\pi} \int_0^\infty |\text{sinc}(\omega/4)|^{\alpha+1} d\omega$ decays only like $\omega^{-(\alpha+1)}$, so it is computed as a finite part plus its analytic tail — Unser & Blu’s own device for their autocorrelation sum — exact at the anchors, 2 at $\alpha = 1$ and $4/3$ at $\alpha = 3$). Reconstruction preview requires `mz_spline` (fractional POCS is a recorded follow-up).

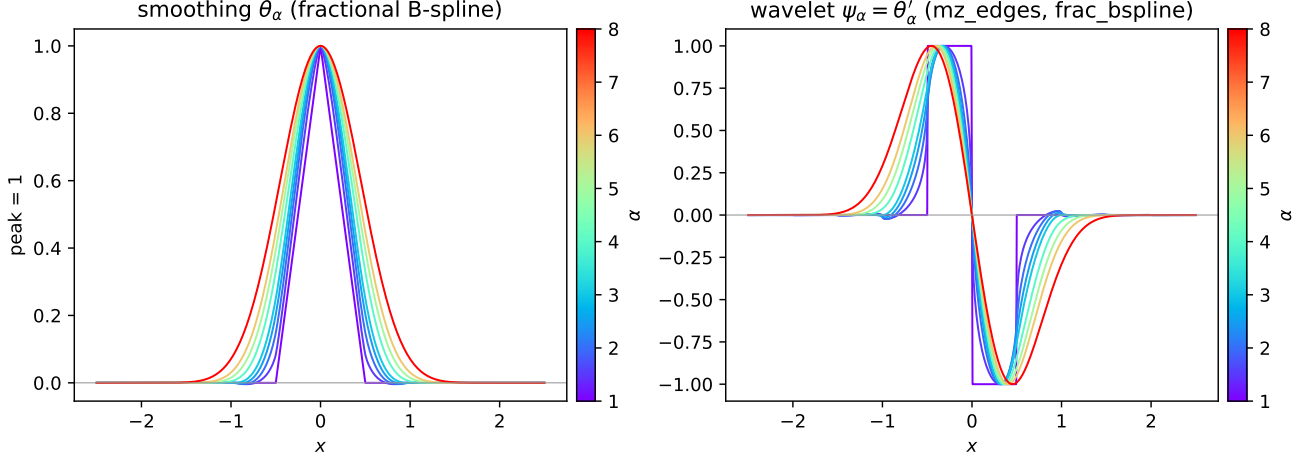


Figure 2: `mz_edges`, `frac_bspline`: the smoothing function $\theta_\alpha(x) = 2\beta_*^\alpha(2x)$, the *symmetric* fractional B-spline of degree α (Unser & Blu 1999, eqs. 10–11; $\hat{\beta}_*^\alpha(\omega) = |\sin(\omega/2)/(\omega/2)|^{\alpha+1}$), and the wavelet $\psi_\alpha = \theta'_\alpha$ as α varies — the analytic limit the filter cascade is pinned to (its maxima sit exactly where the classical $\alpha = 3$ path puts them: no causal phase). α is θ ’s Hölder degree; its approximation order is $\alpha + 1$. $\alpha = 1$: a triangle and a box pair; $\alpha = 3$: the paper’s quadratic-spline wavelet. Only odd integer α gives a compactly supported classical B-spline; other orders decay algebraically, with small sign-changing tails at the knot spacing (the lobes near $x \approx \pm 1$ on the $\alpha \approx 1.5$ – 2.5 wavelets are these, not ringing), and as $\alpha \rightarrow 0$ the symmetric spline is *not* a box but develops logarithmic peaks and oscillating tails (Unser & Blu, fig. 2). Drawn from the exact time-domain series (their eq. 11), never by inverting the Fourier transform, so the $\alpha = 1$ jumps carry no Gibbs ringing. This is Mallat–Zhong’s dyadic *derivative* wavelet built on a fractional spline — exactly one vanishing moment ($\hat{\psi} \sim i\omega$ at low frequency), since maxima-are-edges is the method — not one of Unser & Blu’s fractional spline wavelet bases, which behave like fractional differentiators $|\omega|^{\alpha+1}$.

Parameter	Default	Range	Meaning
<code>n_levels</code> (Levels J)	4	[1, 10]	dyadic levels; scales $2, 4, \dots, 2^J$ px
<code>coarse</code>	full	full / thumbnail	coarse-channel policy (mapping vs coding mode)
<code>dither</code>	off	bool	$\pm \frac{1}{2}$ LSB dither before analysis
<code>wavelet</code>	<code>mz_spline</code>	<code>mz_spline</code> / <code>frac_bspline</code>	the paper’s quadratic-spline bank, or the fractional order
<code>alpha</code> (α)	3.0	[0.1, 8]	fractional spline order (binds under <code>frac_bspline</code> ; 3 = the paper)

3.4 `wavelet_skeleton` — medial-axis (modulus-minima) extractor

Tang–You 2003 (the constructed compact-support wavelet κ : support radius = scale, maxima-pair separation = scale, width- and gray-invariant) + You et al. 2006 (modulus *minima* sit on the medial axis, scale-independently). A detector, not statistics: directional minima gated by a flanking ridge *pair* (each side within reach $\sim s$ must clear `edge_frac` of the peak modulus), valley contrast (`t_frac` of the flanking ridge — the paper’s threshold T made local), and pair dominance (a pair under `pair_dom` of the strongest reachable ridge is someone else’s slope). Border and nodata neighborhoods (one kernel support) are distrusted. On `input=gradient` the detector runs

over $\|\nabla s\|$, where a diffused scarp (an *edge* in the signal) becomes a ribbon whose axis is the fault trace. Output is single-scale extrema (H-lines via `line_id`).

Parameter	Default	Range	Meaning
<code>s1</code> (s_1)	6	[2, 64] px	stage-1 scale = kernel support radius; should cover the ribbon width
<code>s2</code> (s_2)	6	[2, 64] px	refinement-stage scale
<code>t_frac</code> (T/ridge)	0.5	[0.1, 0.9]	keep minima below this fraction of their own flanking ridge
<code>edge_frac</code> (Edge/peak)	0.1	[0.01, 0.6]	flanking-pair significance vs the <i>peak</i> modulus; lower to admit faint features
<code>n_stages</code>	2	[1, 5]	WT passes (1 = detection only; each further pass re-runs on the skeleton image)
<code>input</code>	signal	signal / gradient	ribbons in the signal, or scarps via $\ \nabla s\ $
<code>wavelet</code>	tang_you	tang_you / gaussian / bspline2	the papers' own comparison set; theorems hold for κ only
<code>pair_dom</code> (Pair dom.)	0.5	[0, 0.9]	pair-dominance gate; 0 disables

3.5 cdf_edges — complex cross-diffusion edges

One complex cross-diffusion evolution (Gilboa small- θ ; Barbeiro 2023 for the scale-space reading), snapshotted at the dyadic schedule $n = \sigma^2/(2 \cos \theta)/dt$: the Re channel evolves as the Gaussian-smoothed image and $\text{Im}/\theta \approx t \cdot \text{LoG}$ (machine-pinned), so one evolution *is* a Mexican-hat scale space. Edge modes: `mz` = grad-Re NMS maxima (*our* graft — the M-Z definition applied to a different generator, making `cdf_edges` directly comparable to `mz_edges` at matched scales); `marr` = Im zero crossings (the CDF authors' own edge concept — Barbeiro's *v* “in the role of edge detector”). The nonlinear variant modulates diffusivity by the Im channel (Perona–Malik-style edge stopping through k). Final Re (`filtered`) and Im (`edge_channel`) rasters ride the result. Scales are effective Gaussian σ in px.

Parameter	Default	Range	Meaning
<code>n_levels</code> (Levels J)	4	[1, 8]	dyadic sigmas $1, 2, \dots, 2^{J-1}$ px; cost is quadratic in the top σ
<code>variant</code>	linear	linear / nonlinear	LCDF, or NCDF with edge-stopping
<code>k_edge</code> (k)	1.0	[0.001, 1000]	Perona–Malik edge threshold, image units (nonlinear only)
<code>theta</code> (θ)	0.1047	[0.001, 0.5] rad	complex-diffusion angle (small- θ regime; $\pi/30$ default)
<code>dt</code>	0.2	[0.001, 0.24]	explicit-Euler step (stability-capped)
<code>edge_mode</code> (Edges)	mz	mz / marr	Canny/M–Z maxima of Re, or Marr zero crossings of Im
<code>detector</code>	nms	nms / follow	maxima detector
<code>show</code> (<i>view</i>)	edges	edges / filtered / edge_channel	the edge stack, the final Re, or the final Im
<code>interpolate</code>	off	bool	sub-pixel refinement of the maxima
<code>floor</code>	0.05	[0, 0.5]	detection floor (fraction of the per-snapshot maximum)

3.6 pm_edges — Perona–Malik anisotropic diffusion edges

One Perona–Malik (1990) evolution — real, gradient-driven conduction $g(|\nabla I|)$, $\lambda \leq \frac{1}{4}$, adiabatic borders — snapshotted at the same nominal dyadic σ schedule as `cdf_edges`, with gradient-NMS maxima per snapshot. What it adds: past the flux peak $\phi(s) = s g(s)$ the diffusion runs locally *backward*, sharpening edges while the discrete maximum principle keeps it stable, so edges stay sharp and in place across the stack. Scales are nominal effective σ (the $K \rightarrow \infty$ heat-equation limit); `filtered` (the final diffused raster) rides the result.

Parameter	Default	Range	Meaning
n_levels (Levels J)	4	[1, 8]	dyadic nominal sigmas
k_edge (K)	1.0	[0.001, 1000]	edge threshold in image units: contrasts above K sharpen, below it blur
g	exp	exp / frac	conduction function (the paper’s two)
lam (λ)	0.2	[0.001, 0.25]	explicit step (stability needs $\lambda \leq \frac{1}{4}$)
show (<i>view</i>)	edges	edges / filtered	canvas shows the edge stack, or the diffused raster
detector	nms	nms / follow	maxima detector
interpolate	off	bool	sub-pixel refinement of the maxima
floor	0.05	[0, 0.5]	detection floor (fraction of the per-snapshot maximum)

4 Decompositions

Standalone transforms heading their own chain, displaying through the ordinary raster path. Every output is computed once and cached; which one is on screen is a *view* choice. To keep an output as data of its own, fork it (§1).

Group (**tucker_HOOI_HOSVD**, **ssa2d**; *view*): which components form the reconstruction — **all** (the whole kept reconstruction, exactly), or a list such as 1–3, 5. **recon** shows the sum of the group; **residual** shows the data minus that sum (data = recon + residual of the whole kept reconstruction). So removing just the first component is Group 1 with Show = residual. The layer row names it: **recon** [1–3, 5] (88%), **data** – [1] (42%) (the group’s share). An unreadable group shows the whole reconstruction and says so. The Components window picks the group by clicking thumbnails.

4.1 pca — principal components of a multi-band field

Principal component analysis over the bands of a multi-component field ($n_y, n_x, n_c \geq 2$) (covariance-trick PCA); shows one component as a raster.

Parameter	Default	Range	Meaning
n.components (Components)	6	[1, 64]	components computed
component (Show PC; <i>view</i>)	1	[1, 64]	the component shown
standardize	off	bool	scale each band to unit variance first (correlation PCA)

4.2 tucker_HOOI_HOSVD — delay-embedded Tucker decomposition (Tucker-SSA)

A Tucker decomposition of a delay embedding of the field — Broomhead–King singular-spectrum analysis carried to 2-D and multiband data by a Tucker decomposition instead of one SVD. (It was first named **tucker_havok**; HAVOK’s regression on the delay coordinates was never part of it.) Nothing is ever materialized: bases come from Grams of contiguous window slabs, and the reconstruction averages each pixel over the windows covering it.

- **Embedding.** **delay_2d** (default): *both* axes delayed — every $L \times L$ patch ($L = \text{Delays}$), tensor (delay _{y} , delay _{x} , y' , x' [, band]), so no image axis plays “time”. **delay**: a 1-D Hankel embedding along one axis (Tape axis) — for data where one axis really is time (e.g. a reconstructed tape); on an ordinary 2-D field it biases the result toward that axis. **none**: the array’s own modes (rows, columns[, band]).
- **Method** — a two-state button. **HOSVD**: one pass, each mode’s leading singular vectors. **HOOI**: that HOSVD refined by *HOOI sweeps* alternating passes, each mode re-derived with the others fixed (monotone in the kept core energy; the fast path equals a materialized HOOI to 10^{-8}).

- **Components** (at most 32, largest core-energy share first): under `delay_2d` one per kept (vertical, horizontal) delay-pattern pair (a, b) ; *Orientation combined* adds each pair to its twin (b, a) ($r(r+1)/2$ components for delay rank r). Tape: one per delay pattern; none: one per rows-mode vector. They sum to the reconstruction.
- Separable by construction: a pattern is a vertical pattern $\otimes$ a horizontal one, so an oblique structure spreads over several pairs. `ssa2d` is the rotation-invariant alternative.

Parameter	Default	Range	Meaning
<code>embed</code> (Embedding)	<code>delay_2d</code>	<code>delay_2d</code> / <code>delay</code> / <code>none</code>	see above
<code>axis</code> (Tape axis)	<code>rows</code>	<code>rows</code> / <code>cols</code>	the time axis of the 1-D tape (greyed out unless <code>delay</code>)
<code>n_delays</code> (Delays)	32	[2, 512]	embedding dimension L (patch $L \times L$ under <code>delay_2d</code>); the lag between delays is 1 px
<code>rank_delay</code> (Rank: delay)	4	[0, 512]	delay patterns kept (both delay modes under <code>delay_2d</code>); 0 = full
<code>rank_time, rank_space</code>	0	[0, 10^5]	ranks of the rows/cols (y'/x') modes; 0 = full
<code>rank_band</code> (Rank: band)	0	[0, 64]	rank of the band mode (multiband only); 0 = full
<code>method</code> (Method)	HOSVD	HOSVD / HOOI	the algorithm (a toggle button)
<code>sweeps</code> (HOOI sweeps)	2	[1, 16]	HOOI refinement passes (greyed out under HOSVD)
<code>show</code> (<i>view</i>)	<code>recon</code>	<code>recon</code> / <code>residual</code> / <code>component</code>	what the canvas shows
<code>component</code> (<i>view</i>)	1	[1, 64]	the component shown (Show = component)
<code>pairs</code> (Orientation; <i>view</i>)	<code>separate</code>	<code>separate</code> / <code>combined</code>	component numbering under <code>delay_2d</code> ; also what a Group refers to
<code>group</code> (Group; <i>view</i>)	<code>all</code>	<code>text</code>	the components the recon sums (above)

Superseded: `tucker_havok` stays registered (browser Dev group) so saved projects open and resolve exactly as before; its single *HOOI sweeps* knob chose the algorithm (0 = HOSVD).

4.3 ssa2d — 2D singular-spectrum analysis

Golyandina & Usevich (2010). Every $L_r \times L_c$ window of the image is one column of the Hankel-block-Hankel matrix $\mathbf{W}$ ($L_r L_c \times K_r K_c$, $K = N - L + 1$); *one* SVD $\mathbf{W} = \sum_i \sqrt{\lambda_i} U_i V_i^\top$ gives the eigentriples. Reshaped, U_i is an **eigenarray** — an $L_r \times L_c$ pattern of any shape, not restricted to separable products, which makes 2D-SSA invariant to rotations of the image axes — and V_i a **factor array** ($K_r \times K_c$). Each rank-1 term, averaged back onto the image (every pixel over the windows covering it), is an **elementary component**; all of them sum to the image. Ranks (the paper’s §4, pinned by tests): an oblique cosine $\cos 2\pi(\omega_x k + \omega_y l)$ is rank 2, a product of cosines rank 4, a sum of m 2-D exponentials rank m .

Computed without forming $\mathbf{W}$: the lag covariance $\mathbf{W}\mathbf{W}^\top$ ($(L_r L_c)^2$) accumulated from blocks of window rows, its eigenbasis, then $\mathbf{W}^\top U$ in the same blocks; float64 BLAS, no FFT; equal to an explicit SVD of $\mathbf{W}$ to 10^{-9} . Windows are limited to 1600 entries (40×40 , about 4 s on a 256^2 field; larger ones would need Lanczos with an FFT matvec). Single-band fields only. NaN pixels are filled with the finite mean for the decomposition and re-masked in every output. Also returned: the eigenvalue shares $\lambda_i / \sum \lambda$ and the **w-correlation** matrix (the paper’s Def. 3.18) of the kept components — the grouping aid shown in the Components window.

Parameter	Default	Range	Meaning
<code>rows_window, cols_window</code>	16	[2, 512]	window $L_r \times L_c$ ($L_r L_c \leq 1600$)
<code>n_components</code> (Components kept)	16	[1, 64]	leading eigentriples kept
<code>show (view)</code>	recon	recon / residual / component	what the canvas shows
<code>component (view)</code>	1	[1, 64]	the elementary component shown
<code>group (Group; view)</code>	all	text	the components the recon sums (above)

5 Microcanonical tools

Per-pixel singularity exponents $h(x)$ from the model $T(x, r) \approx \alpha(x) r^{h(x)}$ (Turiel 2008), $D(h)$ by direct histogram — the formalism that is *not* canonical/Legendre. One tool per functional; both expose the Pont 2006 estimator pair: **regression** (per-pixel log-log slope over the scale grid; the prefactor cancels exactly; resolves both limbs of $D(h)$) and **punctual** (the finest scale alone, ensemble-mean normalized; sharpest localization, the most-singular-set extractor; cannot resolve the right limb; carries an $\alpha(x)/\langle\alpha\rangle$ bias). Both estimators share the same cross-scale no-support mask — they agree on *where* an exponent exists and differ in *how*. Under punctual, $\kappa/\mathbf{n_scales}$ shape only the mask and `r2_map` is honestly all-NaN.

5.1 holder_measure — gradient-measure route

T = positive unit-mass kernel over $\|\nabla s\|$ (finite differences first, then g_0 -class smoothing). For measure-like / edge-sparse fields; slope = h . The scale knob r_1 is the kernel’s *dilation scale* (for the Gaussian: its standard deviation) — positive kernels have no zero crossings, and their resolution floor is the pixel, except for the compactly supported q-Gaussian (below). Kernel trade-off (Turiel App. A): Gaussian resolves every h ; Lorentzian truncates at $h < 2\beta - d$ but localizes sharpest; q-Gaussian interpolates; `frac_gaussian` fractionalizes the envelope, $e^{-\rho^n/2}$ ($n=2$ is the Gaussian exactly).

The q-Gaussian is Tsallis’s $e_q^{-\rho^2/2r^2}$ with $e_q^x = [1 + (1 - q)x]_+^{1/(1-q)}$ (Borges et al. 2004, eq. 9), for $-1 \leq q \leq 3$. Above $q = 1$ it has power-law tails (Student-t / Lorentzian: $q \equiv 1 + 1/\beta$ with ρ^2/r^2 rescaled); $q = 1$ is the Gaussian; below $q = 1$ it is *compactly supported*, radius $r\sqrt{2/(1-q)}$ — $q = 0$ is the Epanechnikov paraboloid, $q = -1$ a half-dome of radius r , and $q \rightarrow -\infty$ the Dirac δ . For $q \geq 2$ the 2-D kernel has no finite mass (the Lorentzian $\beta \leq 1$ case), so the result depends on the field size; the q knob shows a warning there. The compact kernel (below $q = 1$) is the steepest decrease there is, so it carries a resolution floor: the r_1 knob warns when its support disc, of radius $r/\sqrt{\beta(1-q)}$ for width β , is less than 2 px across, i.e. below $r_1 = \sqrt{\beta(1-q)}$ (at most 1 px at the default width).

5.2 holder_multiaffine — wavelet-projection route

$T = |\psi_r * s|$ with a zero-mean wavelet applied to the *raw signal* (derivative order lives inside the kernel). For smooth / function-class fields (DEMs); slope = γ , with $h_{\text{measure}} = \gamma - 1$. Here r_1 is the *zero-crossing radius*: every wavelet in the menu is calibrated to change sign at radius r , so for a wavelet with one crossing the fig.-2 minimum-resolution rule reads $r_1 \geq 1$ (more crossings, or a compact support, raise it: *Resolution floor* below). The g-ladder is the vanishing-moment axis (order n measures only $\gamma < n$): **g1/g3** are the fractional family at $n=1, 3$; **g2** is the Ricker; **lorentzian_marr** is the Laplacian (Marr operator) of the Lorentzian envelope (true tail $\rho^{-2\beta-2}$, truncation $\gamma \geq 2\beta$); **frac_gaussian** makes the order a real knob ($\psi \propto {}_1F_1(\frac{n+2}{2}; 1; -\rho^2/2\sigma^2)$, $n=2$ reproduces the Ricker exactly): n is a fractional *differentiation* order ($\hat{\psi} \sim |k|^n$ at low frequency), so exact moments vanish only for the integer orders below n — a fractional n is not “ n vanishing moments” — and $\gamma < n$ is measurable.

`q_mexican` is the q-Mexican hat of Borges et al. (2004, eq. 15): the second derivative of the q-Gaussian *raised to the power* $2 - q$. In 2-D (the Laplacian) this gives

$$\psi_q(\rho) \propto [1 - (2 - q)b\rho^2] [e_q^{-b\rho^2}]^q, \quad b = \frac{1}{(2 - q)r^2},$$

so the zero crossing sits at $\rho = r$ for every q , and $q = 1$ is the Ricker exactly. The paper’s $q < 3$ is the 1-D bound: in 2-D the Laplacian of $[e_q]^{2-q}$ vanishes identically at $q = 2$ and never changes sign above it, so the knob runs $-1 \leq q \leq 1.95$. For $q \geq 0$ the wavelet is bounded; for $q < 0$ it diverges at its cutoff (still integrable) and the knob shows a warning. Because $[e_q]^{2-q}$ is itself a q' -Gaussian with $q' = 1/(2-q)$, this is the family of Laplacians of q -Gaussians under a relabelled index: **q_mexican** at q equals **lorentzian_marr** at $\beta = (2-q)/(q-1)$ for $1 < q < 2$ (e.g. $q = 1.5 \equiv \beta = 1$), with tail $\rho^{-2/(q-1)}$ and truncation $\gamma \geq 2(2-q)/(q-1)$.

The q width β (both q -kernels) is the β of $e_q^{-\beta x^2}$, a toggle away from the paper’s two choices.

- **Fixed** (default $\beta = \frac{1}{2}$): the unit-variance Gaussian per component at $q = 1$, in any dimension. The scale r keeps its meaning at $q = 1$: $\sigma = r$ for the q -Gaussian, the zero crossing at r for the q -Mexican hat. At other q the q -Mexican hat’s zero crossing drifts to $r/\sqrt{2\beta(2-q)}$.
- **q-paired**: Borges et al.’s $\beta = 1/(3-q)$ holds the q -Gaussian’s *escort* (q -)variance $\int x^2[e_q]^q / \int [e_q]^q$ at one. In d dimensions that variance is $1/(\beta(d+2-dq))$ per component, so the pairing is $\beta = 1/(d+2-dq)$: $1/(3-q)$ in 1-D and $\beta(q) = 1/(2(2-q))$ here.
- The q -Mexican hat’s zero crossing sits exactly at that escort RMS radius, so the pairing also pins it (at r , for every q : the tool’s classic calibration). The two settings agree at $q = 1$.
- The pairing exists exactly where the 2-D q -Mexican hat is admissible, $q < 2$: the escort variance and the zero mean both need $d+2-dq > 0$. A q -paired q -Gaussian therefore refuses $q \geq 2$.
- In real space the normalisation is done numerically at every scale, so the paper’s closed-form constants are not used. (In 2-D they happen to be elementary: $A_q = \sqrt{\beta(3-q)(1+q)}/\pi$ for the q -Mexican hat, $B_q = \sqrt{\beta(3-q)}/\pi$ for the modulated q -Gaussian. The flat Jacobian $\rho d\rho = du/2\beta$ turns every integral into a Beta function with integer arguments, where 1-D keeps $\Gamma(\frac{1}{2})$ -type ratios.)

Resolution floor on r_1 . A discretized wavelet must separate its positive and negative parts, so its zero crossings set the minimum attainable resolution, and more crossings cost resolution (Turiel et al. 2008, fig. 2 and §4.2.1); Turiel (2009) puts a Mexican hat’s inner minimum scale at “several pixels”. Neither gives a number. The tools read the rule as: every sign lobe at least 2px across at r_1 , the central disc counted by its diameter. The edge of a compact support is not a sign change, but it closes the last lobe. The r_1 knob warns below the floor and never refuses.

- One crossing at r (**g1**, **g2**, **lorentzian_marr**, **frac_gaussian** for $n \leq 2$): the floor is $r_1 = 1$, as before.
- More crossings: **g3** 1.32px (its ring is 1.52 r wide); **frac_gaussian** at $n = 4$ and 5, 1.41 and 1.46px.
- **q_mexican**: its crossing sits at $\rho_0 = r/\sqrt{2\beta(2-q)}$ (r when q -paired) and, below $q = 1$, its support ends at $\rho_0\sqrt{(2-q)/(1-q)}$. The negative ring between them narrows as q falls, and below $q \approx 0.875$ (q -paired) it is the narrowest lobe: floor 2.73px at $q = 0.5$ and 4.42px at $q = 0.1$. A fixed width β is a pure dilation of the wavelet, so it rescales the floor by $\sqrt{2\beta(2-q)}$.
- **holder_measure**: only the compact q -Gaussian has a floor (above).

Wavelet space: real or Fourier (both tools). *Real* (the default, Turiel’s construction) samples each kernel on the pixel grid and normalizes it there. *Fourier* evaluates the same continuous kernel’s closed-form 2-D transform at $r\mathbf{k}$: a Matérn function of k for a q -Gaussian with $q > 1$ (and the Lorentzian, $q = 1 + 1/\beta$), a Jahnke–Emde function for $q < 1$, $k^n e^{-k^2/4u_0}$ for the fractional family, evaluated stably at every q . The kernel is then band-limited rather than sampled.

- Measure route: unit mass exactly. Multiaffine route: zero mean exactly, and the kernel is the L^1 -dilated continuous wavelet, so T equals the real construction’s times the mother wavelet’s L^1 norm (a constant that cancels in every slope) wherever both are exact.
- They differ where sampling matters: at the smallest scales, for compact kernels (the Fourier one rings slightly), and for heavy tails, which real space cuts at the edge of the padded grid (about 10^{-4} at the default padding for a $\rho^{-6.7}$ tail).

- Fourier is refused on the measure route for `frac_gaussian` (no closed-form transform), and for a q-Gaussian at $q \geq 2$ or a Lorentzian with $\beta \leq 1$ (no finite mass to normalize by).

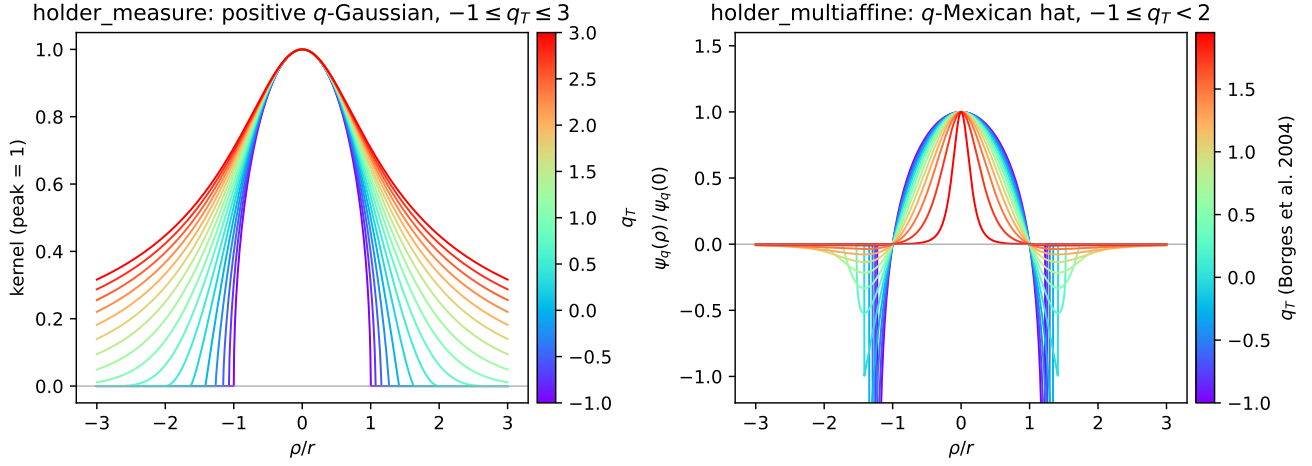


Figure 3: The q family. Left: `holder_measure`'s positive q-Gaussian, $-1 \leq q \leq 3$ (compact below $q = 1$). Right: `holder_multiaffine`'s q-Mexican hat (Borges et al. 2004, in 2-D), $-1 \leq q < 2$; every curve crosses zero at $\rho = r$, and for $q < 0$ the negative lobe diverges at the cutoff. Profiles scaled to 1 at the centre.

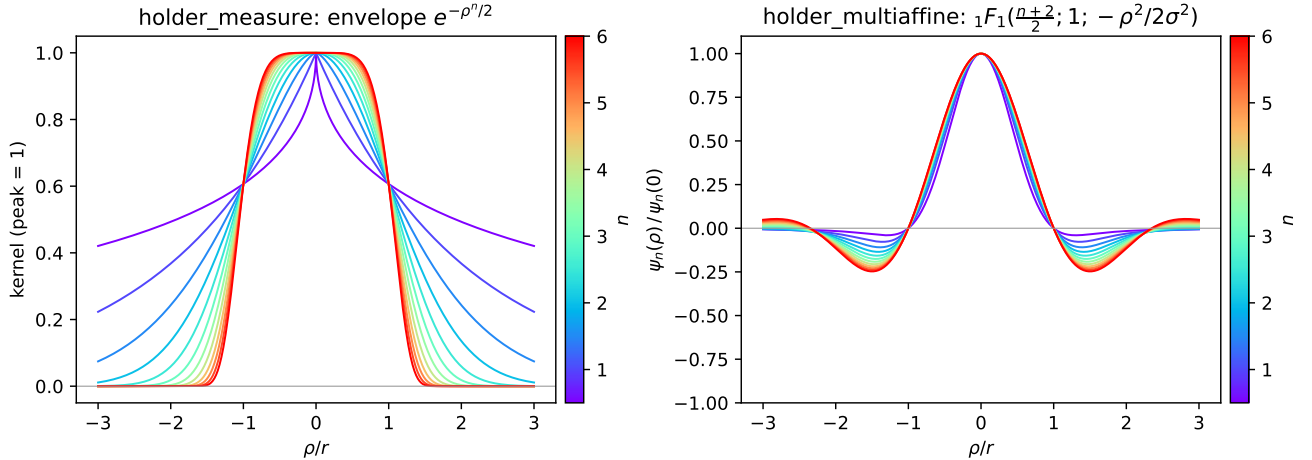


Figure 4: The fractional Gaussians. Left: `holder_measure`'s envelope $e^{-\rho^n/2}$ ($n < 2$ fattens the tail, $n > 2$ flattens the top; every curve passes $e^{-1/2}$ at $\rho = r$). Right: `holder_multiaffine`'s order- n wavelet ${}_1F_1(\frac{n+2}{2}; 1; -\rho^2/2\sigma^2)$, σ set per order so the first zero crossing sits at $\rho = r$.

The same r_1 value is not numerically comparable across the two tools, and γ values are not comparable across wavelets.

Parameter	Default	Range	Meaning (both holder tools)
estimator	regression	regression / punctual	Pont estimator pair (see above)
r_min (r_1)	1.0	[0.1, 32] px	finest scale (dilation scale / zero-crossing radius per tool); warns below the resolution floor
kappa ($\kappa = r_2/r_1$)	8	[1.1, 200]	scale-range width; the knob that changes the physics
n_scales	6	[2, 64]	samples in the range (nearly irrelevant vs κ ; keep small)

wavelet q_tsallis (Tsallis q , q_T)	(per tool) 1.5	method-true menu [−1, 3] / [−1, 1.95]	see tool text q-family shape, measure / multiaffine (q-gaussian / q-mexican only); warns at measure $q \geq 2$, multiaffine $q < 0$
beta (β)	1.0	[0.1, 8]	Lorentzian tail exponent (Lorentzian family only)
frac_n (n) q_pairing (q width)	2.0 fixed	[0.05, 8] fixed / q-paired	fractional order (frac-gaussian only) the q-kernel’s width β : fixed, or $\beta(q) = 1/(2(2 - q))$ (below); greyed out unless q-gaussian / q-mexican
q_beta (q β)	0.5	[0.05, 5]	the fixed width of $e_q^{-\beta \rho^2 / r^2}$ (greyed out while q-paired)
construction (Wavelet space)	real	real / fourier	sample each kernel on the grid, or build it from its closed-form transform (above)

5.3 band_recon_measure / band_recon_multiaffine — reconstruction from an h band

The same $h(x)$ engine as the matching holder tool (its params verbatim, shared by identity) + the singularity band $h \in [h_{lo}, h_{hi}]$ and an island sieve, then gradient-restricted Fourier reconstruction (Turiel eq. 65, symmetric-extension border fix). Sweeping the band and watching the reconstruction *is* the workflow; the Band dialog commits the variant matching the layer’s producer.

Parameter	Default	Range	Meaning (band knobs; engine knobs above)
h_lo / h_hi	−2 / 0	[−6, 6]	the band $h_{lo} \leq h < h_{hi}$, in the estimator’s own frame
min_island / max_island	0 / 0	px	drop mask components outside this size range before inversion
connectivity	8	8 / 4	island connectivity (8 keeps diagonal ribbons whole)

Superseded: **holder_map** and **band_recon** (the pre-split conflated tools, estimator $\times$ wavelet crossed) remain registered for saved projects and live in the browser’s Dev group; their physics is identical to the split tools at the old defaults.

6 Topology

6.1 chain_topology (Transform)

Builds the incidence graph over the WTMM maxima-line skeleton (no parameters). Must follow **wtmm2d**; downstream topology filters query the graph it builds.

6.2 min_vchains (Filter)

Keeps H-line points that at least **min_vchains** maxima lines (V-chains) thread. Default UI opens at 1 (show everything computed; turn up to prune).

7 Extrema filters

Act on the per-scale extrema layers of any producing analyzer (**wtmm2d**, **mz_edges**, **wavelet_skeleton**, **cdf_edges**).

Device.param	Default	Range	Meaning
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<code>scale.select.scale_idx</code>	0	[0, 255]	pick ONE scale from the stack — <i>the scale scrubber</i> ; a list index, instant r
<code>orientation.wedge.centre</code>	0	deg	keep extrema whose gradient strike f
<code>orientation.wedge.half_width</code>	90	[0, 90] deg	wedge
<code>orientation.wedge.north</code>	grid	grid/true/magnetic	wedge half-width (90 = no-op)
<code>modulus.threshold.frac</code>	0	[0, 1]	the NorthRef of centre ; frame error
<code>hline.length.min_len/max_len</code>	1 / 0	points	multi-degree errors
<code>hline.modulus.frac</code>	0	[0, 1]	per-point: keep $ W \geq \text{frac} \times \text{scale}$ p
<code>hline.modulus.mode</code>	sup	sup / sup_all_scales / mean / segments	per-line point count range; kills orph
			confetti and stitching seams; -1 orph
			per- <i>line</i> modulus keep-fraction
			line statistic: peak at this scale; peak
			threading chains (needs chains); mea
			per-segment between modulus minim

8 Chain filters (WTMM maxima lines)

Act on cross-scale chains — `wtmm2d` results only (the other analyzers stamp `chains:` `[]`; their cross-scale linking is reserved work).

Device.param	Default	Range	Meaning
<code>chain_holder.cutoff</code>	-3	[-3, 3]	keep chains with Hölder $\geq$ cutoff
<code>chain_holder.estimator</code>	ols	ols / max	per-chain h : log-log OLS slope, or the max-modulus scaling read
<code>chain_modulus.threshold</code>	-99	[-99, 32]	keep chains whose max $\log_2 W \geq$ threshold
<code>chain_length.min_len</code>	2	[2, 64]	minimum scales spanned

8.1 chain_classify

Classifies chains by their Hölder exponent and geometry into step-like ($h \approx 0$: fault-scarp class) vs delta-like ($h \approx -1$: thin-ridge class) lineaments, with a straightness gate; flagged or excluded per **action**, excluded chains optionally drawn as ghosts.

Parameter	Default	Range	Meaning
<code>estimator</code>	ols	ols / max	per-chain h estimator
<code>fit_a_min</code>	2.0	[0.25, 16]	fit floor: scales below this are excluded from the h fit
<code>h_step_lo/hi</code>	-0.15 / 0.15	[-3, 3]	the step-class h window
<code>h_delta_lo/hi</code>	-1.2 / -0.8	[-3, 3]	the delta-class h window
<code>straightness_min</code>	0.85	[0, 1]	end-to-end / arc-length straightness gate
<code>action</code>	flag	flag / exclude	tag matches, or drop non-matches
<code>show_ghosts</code>	on	bool	draw excluded chains desaturated

9 Groups, points, dev

`group.paint` / `group.filter`: the arrangement view's commit transaction — `group.paint` stamps committed group tags onto chain copies (`spec_json/signature` are written by the commit flow, not hand-edited); `group.filter` keeps or drops one named group.

`backproject`: registers a point catalogue onto a raster layer's grid (the point-layer transform). **target** names the raster layer; the `_target_*` params are the resolved grid geometry, written by the shell — not hand-edited.

derived_vectors: a derivative dataset’s frozen extrema and maxima lines, read from its own file (§1). Placed by the shell (browser Dev group); its one parameter (the file) is hidden.

stub_wavelet / stub_holder / stub_wedge: parameter-schema test fixtures (the browser’s Dev group); no analysis role.

10 Planned: q-generalized wavelets

Not implemented yet — recorded here so the design stays in one place. Borges et al. (2004) generalize the Mexican hat (above), the Morlet wavelet (a q-Gaussian-modulated plane wave, their eq. 23) and the trigonometric functions (an even/odd pair of q-trigonometric wavelets, their §4), and note that “other wavelets, as well as scaling functions, may be generalized along the lines here introduced, for instance, the Meyer wavelet or the sinc function.”

- **q-generalized g1 and g3 for wtmm2d**. WTMM’s analyzing wavelets must stay *derivatives of a smoothing function* ($\psi = \partial\theta$), so that maxima mark edges. The filters are built in the Fourier domain, so they need the 2-D transform of the q-Gaussian, a Hankel transform with Bessel order $1/(q-1) - d/2$, one half lower than the paper’s 1-D order. That transform is now built and evaluated stably at every q (the Matérn and Jahnke–Emde functions used by *Wavelet space: Fourier*, §5).
 - **q-g1**: $\partial\theta_q$ with θ_q the unit-mass q-Gaussian, so the L^1 normalization is unchanged ($\hat{\theta}_q(0) = 1$); it exists for $q < 2$.
 - **q-g3**: the power- $(2-q)$ recipe makes the Laplacian’s envelope a q' -Gaussian with $q' = 1/(2-q)$, whose mass is finite only for $q < 3/2$. For $3/2 \leq q < 2$ the wavelet is still finite (k^2 cancels the divergence at $k = 0$), but $\hat{\theta}(0) = 1$ can no longer fix its scale, so it needs another normalization there, such as unit L^1 norm.
- **q-Morlet** and the **q-trigonometric** pair (oriented/directional analysis).
- **q-Meyer** and **q-sinc**, as the paper suggests.

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Appendix: result contracts

- **Extrema schema** (all edge/skeleton analyzers): per-scale `{x, y, mod, arg, line_id}`; `line_id` groups H-lines, `-1` marks isolated points. `scales` is the matching scale array (px).
- **h-map contract** (holder tools): `h_map` displays in place of the field through the ordinary raster path; `r2_map` rides along (all-NaN under punctual). `raster_out` (band tools) displays the reconstruction; `h_map/band_mask` ride for the dialogs.
- **cdf extras**: `filtered` (final Re) and `edge_channel` (final Im) rasters.
- The vector (3-D) view draws the *finest* extrema layer only — put `scale_select` in the chain to choose which.
- **Decomposition extras**: `ssa2d` — `ssa_recon`, `ssa_residual`, `ssa_components` (n, n_y, n_x) , `ssa_eigen_share`, `ssa_eigenarrays` (n, L_r, L_c) , `ssa_factor_arrays` (n, K_r, K_c) , `ssa_w_correlation` (n, n) . `tucker_HOOI_HOSVD` — `tucker_recon`, `tucker_residual`, `tucker_components` / `tucker_combined_components` with their energies, core, factors, energy per mode; on a multiband field also `tucker_recon_bands` / `tucker_residual_bands` (n_y, n_x, n_c) (the display shows band 1). A view step writes `raster_out` and, for a named view, `_view_note` (what the layer row appends).
- **Derivative file**: the RasterField npz keys (`schema`, `name`, `values`, `x_axis`, `y_axis`, `units`, `frame_meta`, `provenance`) plus, when it carries vectors, `vec_e*` (extrema, CSR-packed), `vec_c*` (maxima lines) and `vec_scales`. A vector-only derivative's raster is all-NaN on the same grid.